Deep Learning-Based Monocular Visual Odometry for Autonomous Spatial Mapping

Interim report on

Deep Learning Project

[ICT-4442]

Submitted By

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PART B : INTERIM REPORT

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Title of the Project: Deep Learning-Based Monocular Visual Odometry for Autonomous Spatial Mapping

GitHub Repository Link: https://github.com/VaibhavBansal05/Deep_Monocular_VO

1. Literature Review

The nine reviewed papers cover the main ideas used in our four models: classical visual odometry, supervised learning, self-supervised learning, recurrent networks and attention.

Virtual KITTI 2 recreates scenes from the Karlsruhe Institute of Technology and Toyota Technological Institute benchmark (KITTI). It provides red-green-blue (RGB) images, depth and exact six-degree-of-freedom (6-DoF) camera poses.

The table uses the following common terms: convolutional neural network (CNN), long short-term memory (LSTM), gated recurrent unit (GRU), convolutional GRU (ConvGRU), vision transformer (ViT), reinforcement learning (RL), graphics processing unit (GPU) and light detection and ranging (LiDAR).

Paper (Author, Year)	Method	Dataset	Key Result	Relevance to Project
Cabon et al., 2020 — Virtual KITTI 2	Synthetic copies of five KITTI sequences with weather and camera changes; includes RGB, depth, motion, segmentation and camera poses	Virtual KITTI 2 (synthetic) versus real KITTI	Synthetic data is useful for training and testing, but it still differs from real data	Provides the synthetic images, camera data and exact poses used in this project
Wang et al., 2017 — DeepVO	A CNN extracts image features and a two-layer LSTM predicts camera motion	KITTI odometry	Reached performance close to classical methods and learned metric scale from data	Provides the main architectural basis for Model 2
Zhou et al., 2017 — Structure-from-Motion Learner	DepthNet and PoseNet learn from the difference between a real image and a reconstructed image	KITTI odometry / Eigen split	Outperformed a short-sequence classical baseline but not its full-sequence version	Provides the starting point for Model 3's self-supervised training
Godard et al., 2019 — Monodepth2	Improved the image-reconstruction loss, multi-scale training and handling of static pixels	KITTI (Eigen split)	Reported 0.115 absolute-relative depth error at 640x192 resolution	Guides Model 3's photometric loss and planned ablation studies
Zhao et al., 2019 — Learning Kalman Network for Visual Odometry	Learns observation and motion models, then combines them with a Kalman update	KITTI	Learns metric scale and filter parameters from data	Provides an alternative to the recurrent design used in Model 2
Zhi et al., 2022 — Multi-Scale Unsupervised Monocular Visual Odometry	Uses multi-scale convolutions and self-attention to capture local and long-range motion	KITTI	Improved rotation estimates, especially in scenes with large turns	Supports the attention-based design tested in Model 4
Albukhari et al., 2025 — MINI-DROID-SLAM (Sensors)	Replaces DROID-SLAM's ConvGRU with a smaller GRU while keeping depth and optimisation	TartanAir	Similar accuracy with lower training time and cost	Evidence for Model 2's recurrent-unit ablation
Xie et al., 2025 — CoProU-VO	Uses a ViT and an uncertainty mask to reduce the effect of unreliable image regions	KITTI, nuScenes	Outperformed earlier unsupervised two-frame methods, especially on highways	Suggests uncertainty modelling as a possible extension to Model 3
Messikommer et al., 2024 — Reinforcement Learning Meets Visual Odometry	An RL agent chooses keyframes and grid size inside a classical visual-odometry system	Public VO benchmarks	Improved accuracy and reduced manual tuning	Supports Model 1's combination of classical geometry and learned regression

Synthesis.

Early work tried to learn the complete visual-odometry pipeline. Later studies often kept the useful geometric parts and learned only selected components. Our four models cover both approaches.

Supervised models learn metric scale from pose labels. Self-supervised monocular models usually recover motion only up to an unknown scale. For this reason, we report trajectory error with and without correcting one global scale factor.

The reviewed attention methods also change other parts of the network, so they do not show the effect of attention alone. Our Model 4 tests will keep everything else fixed and change only the attention module. Published KITTI scores are not directly comparable with our Virtual KITTI 2 results because the datasets and test protocols differ.

2. Dataset and Preprocessing

Acquisition.

Virtual KITTI 2 provides separate archives for RGB images, depth and text ground truth. The current working subset contains RGB, intrinsics and poses; dense depth has not yet been added. Our subset contains 8,642 RGB frames from 18 sequences. We checked the extracted frame counts against the dataset documentation.

Preprocessing pipeline.

- Resize. Images are resized from 1242x375 to 128x416 pixels.
- Camera parameters. Focal lengths and principal-point coordinates are scaled to match the resized image.
- Normalization. Images use the same ImageNet normalization as the pretrained 18-layer residual network (ResNet-18).
- Frame groups. Models 1 and 4 use two frames. Models 2 and 3 use five frames.
- Pose conversion. World-to-camera poses are converted into motion between consecutive frames. Camera 0 is selected because both cameras are stored in the same file.
- Augmentation. All frames in a group receive the same colour change and horizontal flip. The seed, epoch and sample number control the random choices, making reruns reproducible.

Split.

The split is by scene, so no geometric content is shared between train and test. Train: Scene01, Scene06, Scene18 under clone, morning, overcast and sunset (4,212 samples). Validation: Scene02/clone (232). Test: Scene20/clone (836). The rain, fog, 15-degree and 30-degree variations are held out entirely for robustness evaluation. All four models use this split. Automated checks confirm that no scene appears in more than one partition.

Verification.

We rebuilt the ground-truth trajectory from the relative poses. It matched the original trajectory to numerical precision, which confirms that the pose direction and camera coordinates are handled correctly.

3. Models and Interim Results

Protocol.

Models 1, 2 and 4 use the same data split and random seed (42). Model 1 (M1) combines classical feature matching with a multilayer perceptron (MLP). Model 2 (M2) uses ResNet-18 followed by a two-layer long short-term memory network. Model 4 (M4) uses ResNet-18 with a Convolutional Block Attention Module (CBAM). M2 and M4 were trained for 60 epochs using 32-bit floating-point (FP32) arithmetic. The best validation checkpoint from each model was tested once on Scene20/clone.

Absolute trajectory error (ATE) measures the difference between the full predicted and true paths. Similarity transformation alignment, Sim(3), corrects one overall scale factor. Rigid-body transformation alignment, SE(3), keeps the predicted scale. Relative pose error (RPE) measures frame-to-frame motion. The t_rel and r_rel columns show translation and rotation drift. Lower values are better.

Model	Owner	ATE Sim(3) (m)	ATE SE(3) (m)	RPE trans (m)	RPE rot (deg)	t_rel (%)	r_rel (deg/100 m)	Model size (M)	Latency (ms/frame)
M1 - classical grid descriptor + MLP	Pranav Joshi	24.209	68.320	0.388	0.218	36.61	8.40	0.27	9.73
M2 — CNN + 2-layer LSTM	Abhishek Laxman Jadhav	61.646	82.835	0.347	0.251	54.52	19.76	15.39	35.80
M4 — pairwise CNN + CBAM	Vaibhav Krishna Bansal	25.471	75.273	0.364	0.196	35.67	7.89	11.23	17.08

Result interpretation.

M1 gives the lowest full-trajectory error. It is also the smallest and fastest model. M2 gives the lowest frame-to-frame translation error, while M4 gives the lowest rotation error and drift. M1 overestimates forward movement by an average of 0.266 metres per frame. This small repeated error builds up over the complete route.

M4 held-out robustness.

Sequence	ATE Sim(3) (m)	ATE SE(3) (m)	RPE trans (m)	RPE rot (deg)	t_rel (%)	r_rel (deg/100m)
Scene20/rain	44.275	47.441	0.291	0.301	33.53	12.21
Scene20/fog	29.919	31.878	0.319	0.286	24.97	8.99
Scene20/15-deg-left	55.792	105.589	0.447	0.244	43.10	13.08
Scene20/30-deg-right	189.036	190.757	0.523	0.446	62.61	41.60

Rain and fog increase the error, but camera rotation causes a much larger drop in performance. The 30-degree-right sequence reaches 189.0 m Sim(3) ATE. We still need to compare CBAM with no attention and Squeeze-and-Excitation (SE) attention before deciding how much the attention module helps.

Implementation status.

M1 is complete. It detects up to 1,000 Oriented FAST and Rotated BRIEF (ORB) features. FAST means Features from Accelerated Segment Test, and BRIEF means Binary Robust Independent Elementary Features. An 8 by 8 grid summarises horizontal motion, vertical motion and reliable matches, and a Multilayer Perceptron (MLP) predicts the camera movement. M2 is complete and evaluated. It uses ResNet-18 with a two-layer Long Short-Term Memory (LSTM) network to learn motion from five-frame sequences. M4 is also complete and evaluated. It uses a pairwise ResNet-18 model with Convolutional Block Attention Module (CBAM) attention. M3, the self-supervised PoseNet and DepthNet model, is still under development and needs the depth-image archive for evaluation.

Shared infrastructure.

All four models use the same data loader, training structure and evaluation code. This keeps the data split, pose format and metrics consistent. Fixed seeds make the runs repeatable, and automated tests check the data, geometry, models and metrics.

Geometric pose objective.

M1, M2 and M4 calculate translation and rotation errors separately. Translation uses straight-line distance in metres. Rotation uses the smallest angle between the predicted and true orientations, called the geodesic angle on the three-dimensional special orthogonal group (SO(3)). Each part is scaled so that one does not dominate the loss. A small extra term penalises a repeated positive or negative error in any direction.

4. Group Contribution and Individual Responsibilities

This is a jointly developed project. All members contributed to planning, technical discussion, integration review, experiment interpretation and report preparation. The table shows the main area handled by each member.

Member Name	Reg. No.	Primary and Shared Responsibilities
Vaibhav Krishna Bansal	230911546	Led Model 4: ResNet-18 pose model, CBAM integration, training and robustness tests. Also coordinated model integration, reproducibility work and the interim report.
Pranav Joshi	230953248	Led the data pipeline and Model 1: dataset preparation, image and camera-parameter processing, ORB matching, grid descriptor, feature cache, MLP regressor, tests and evaluation.
Abhishek Laxman Jadhav	230953360	Led Model 2: ResNet-18 with LSTM and GRU sequence models. Also developed pose geometry, trajectory reconstruction, image projection and related tests.
Suryansh Bajaj	230953250	Leads Model 3 and evaluation: self-supervised PoseNet and DepthNet, trajectory alignment, ATE, RPE, drift and depth metrics, result comparison and tests.

5. Risks and Remaining Work

Remaining model.

Model 3 is the only proposed model that has not yet been completed. Its projection and warping code is ready, but the depth-image archive is still needed for evaluation. M1 is complete. Its next steps are robustness testing and reducing forward-motion bias.

Risk	Assessment and mitigation
Model 3 training	High impact, moderate likelihood. Photometric self-supervised training can be unstable. We will start with an ImageNet-pretrained encoder. If needed, we will use fewer image scales or a lower resolution.
M1 bias and robustness	M1 is small and accurate, but its +0.266 m average forward error accumulates. We will inspect weak matches, tune the bias penalty on validation data, and test rain, fog and rotated cameras.
Overfitting	Validation error stops improving while training loss continues to fall. The best checkpoints were epoch 27 for M2 and epoch 18 for M4. We will use validation ATE and early stopping.
Camera-angle changes	M4 degrades when the camera is rotated by 30 degrees. We will add camera-angle variation during training and repeat the test for all models.
Effect of attention	M4 differs from M1 and M2 in more than its attention module. We will compare three otherwise identical M4 models: CBAM, SE attention and no attention.
Numerical stability	Mixed precision produced non-finite geometric-loss gradients on the Kaggle NVIDIA T4 GPU. The current runs use FP32, and training checks the gradients before each update.

Timeline to Final submission.

Window	Planned work
2 Oct - 8 Oct	Finish M1 reporting, add the depth dataset, complete the M3 training pipeline, and run the remaining M4 attention tests.
9 Oct - 16 Oct	Train and evaluate M3. Test M1 under weather and camera changes. Run the remaining M2 comparisons.
17 Oct - 23 Oct	Run the final selected models on the same hardware and prepare the combined comparison and error analysis.
24 Oct - 31 Oct	Prepare the final report, presentation, repository release and viva.